

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)			Indicative content • both ions are free to move in molten state • Al^{3+} ions attracted to negative electrode / move towards negative electrode because opposite charges attract • Al^{3+} ions gain three electrons forming aluminium (atoms) • Al^{3+} ions are reduced • $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$ • O^{2-} ions attracted to positive electrode / move towards positive electrode because opposite charges attract • O^{2-} ions lose two electrons forming oxygen (atoms) • two O^{2-} ions lose (four) electrons forming oxygen (molecules) • O^{2-} ions are oxidised • $2\text{O}^{2-} - 4\text{e}^- \rightarrow \text{O}_2$ 5-6 marks Clear understanding of movement of ions and gain/loss of electrons; good attempt at equations <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i>	6			6		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				3-4 marks Some understanding of movement of ions and gain/loss of electrons <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks Basic knowledge of ions moving to electrodes of opposite charge or ions gaining/losing electrons <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
	(b)			award (1) for factor and (1) for relevant explanation e.g. coastal location (1) ⇒ to import ore / bauxite / aluminium oxide (1) near a power station (1) ⇒ process uses huge amounts of electricity (1) do not credit more than one factor	2			2		
				Question 9 total	8	0	0	8	0	0